

Topic 1.14 - Function Model Construction and Application

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Construct linear, quadratic, polynomial, rational, and piecewise models from stated conditions or data.
- Use models to solve equations, compare rates, and interpret parameters and outputs in context.
- Validate a model against conditions, domain, residual evidence, assumptions, and practical constraints.

Section A: Foundation and representation

Question 1

Linear model

No calculator

Construct the linear model through $(2, 11)$ and $(7, 31)$ and interpret its slope if x is hours and y is liters.

Question 2

Quadratic

No calculator

Construct the quadratic with vertex $(4, -3)$ through $(1, 15)$.

Section A continued

Question 3

Factored model

No calculator

Construct a quadratic with zeros -2 and 6 and passing through $(0, 24)$.

Question 4

Polynomial

No calculator

Construct the least-degree polynomial with zeros -1 simple and 3 double and with $P(0) = -18$.

Question 5

Rational

No calculator

A rational model has horizontal asymptote $y = 50$, vertical asymptote $x = -4$, and passes through $(0, 70)$. Construct $R(x) = a/(x + 4) + 50$.

Section B: Application and reasoning**Question 6**

Piecewise

No calculator

A service costs 30 riyals for the first 5 units and 4 riyals for each additional unit, up to 20 units. Construct a continuous piecewise model.

Question 7

Application

No calculator

Using $M(t) = 80 + 500/(t + 5)$, solve $M(t) = 100$ and interpret the result if t is years.

Question 8

Rate

No calculator

For the arch $h(x) = -0.4x(x - 10)$, find the average rate of change from $x = 1$ to $x = 4$ with units of meters per horizontal meter.

Section C: Challenge and error analysis**Question 9**

Parameters

No calculator

In $R(t) = L + A/(t + b)$, interpret L and explain the effect of increasing $|A|$ when $t + b$ keeps the same sign.

Question 10

Validation

No calculator

A quadratic regression predicts 12.4, 18.1, 21.7 where observations are 12.2, 18.3, 21.6. Compute residuals and briefly validate the local fit.

Question 11

Challenge

No calculator

Give two different function families that can pass through $(0, 1)$ and $(1, 2)$. Explain why two points do not uniquely determine a general model family.

Question 12

Critique

No calculator

A model predicts -14 customers at a future time. Explain how to respond mathematically and contextually.

Section D: AP-style multiple choice

MCQ 1

No calculator

Which form is most efficient for a quadratic with known zeros 2 and 7?

- A. $a(x - 2)(x - 7)$
- B. $a(x - h)^2 + k$
- C. $mx + b$
- D. $a/(x - h) + k$

MCQ 2

No calculator

A model $M(t) = 60 + 240/(t + 3)$ has long-run value

- A. 3
- B. 60
- C. 80
- D. 240

MCQ 3

No calculator

Which condition is necessary after solving a piecewise equation?

- A. The answer is positive
- B. The answer lies in the interval for the branch used
- C. Every branch has the same slope
- D. The graph is a polynomial

MCQ 4

No calculator

A quadratic has vertex $(2, 5)$ and passes through $(4, 13)$. What is its leading scale?

- A. -2
- B. 1
- C. 2
- D. 4

AP-style free response 1

Calculator permitted - 6 points

The measured height of an arch is shown below.

x	0	2	4	6	8	10
$h(x)$	0	6.4	9.6	9.6	6.4	0

Distances are in meters.

A. Use quadratic regression or structural reasoning to construct a model.

B. Determine and interpret the maximum height and its location.

C. Find the average rate of change from $x = 2$ to $x = 8$.

D. Explain one limitation of applying the model beyond $0 \leq x \leq 10$.

AP-style free response 2**No calculator - 6 points**

A quantity is modeled by $M(t) = L + A/(t+b)$ for $t \geq 0$. It approaches 80 as t increases, and $M(0) = 180$, $M(5) = 130$.

A. Determine L, A, b and write the model.

B. Solve $M(t) = 100$.

C. Find and interpret the average rate of change from $t = 0$ to $t = 5$.

D. State the algebraic restriction and a contextual assumption needed for future use.
